

Privacy Preserving Distributed Control of a 2-DOF Non-collocated Vibrating System Under Earthquake Excitation

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Abstract— In this work, privacy in cloud computing distributed control is studied. More precisely, we focus on encrypted PID controller for a physical system where the control actions are distributed between cloud servers. Most of the control systems rely on data that often contain privacy sensitive information. Smart power networks, smart water grids, smart buildings, robot swarms, and intelligent transportation systems are some examples of using these control algorithms to achieve linked optimization and automation. The secure multiparty computation using Shamir's secret sharing scheme is investigated as a secure approach where private data is split into computational shares. Since shares reveal no information about private data, computing on shares can be done by servers with low risk. A 2-DOF non-collocated vibrating system under earthquake excitation is going to be controlled by the proposed secure control algorithm integrated into the Newmark method while ensuring correct results and no leakage of information.

Keywords— Digital Privacy, Multiparty computation, Secret sharing, Networked control system, Real-time computation.

I. INTRODUCTION

In recent years, embedded devices and systems become ubiquitous in our environment which makes it possible to collect a vast amount of data. Technologies have been developed in such a way that devices across different functionalities can communicate and make independent decisions for optimizing their performances. Online control algorithms are needed to achieve these goals and this control and monitoring depend on the exchange of data. Data include private information of data owners utilized to reduce the resources for consumption, waste, and overall costs. Although cloud-based and distributed control schemes are powerful, flexible, and efficient, when data is released to third parties in exchange for computed control actions, they increase the risk of cyberattacks such as spying or sabotage. In fact, the use of the public communication network and the data processing on third-party platforms promote eavesdropping and manipulation. Future control schemes should overcome those threats and ensure confidentiality and integrity of the involved data to ensure safety in today's data-driven community. For instance, consider a smart meter that collects fine-grained measurements of the electricity consumption of a household or pressure control in water networks as examples of control systems with the ability to reducing resource losses by accurately predicting the consumption through measurements. Another example

could be Intelligent Transportation Systems (ITS) which will be controlled by cloud computing in the future considering more data exchange to make a unified vehicular network. In this way, data play an important role in these schemes, while revealing private information.

Cryptographic algorithms from ancient cryptography to modern approaches such as symmetric or public-key cryptography were developed substantially for static data, like databases. Plenty of research were dealing with this subject and noticeable improvements have been achieved [1]. Nevertheless, dynamical systems are iterative operations that generate structured and dependent data. Furthermore, system behavior at a specific time is associated with the previous outputs of the system. Hereupon, special attention is needed when using cryptographic techniques in solving optimization problems and implementing control schemes [2].

Regarding the problem, several studies have been carried out, but most of them have faced the problem by the application of differential privacy or secure communication channels [3-5]. Though, these methods are useful, they rely on trusted third-party and data owners are needed to expose their raw data to probably corrupted servers that are susceptible to adversarial attack. Information-theoretic privacy is studied in [6] which also mentions differentially-private mechanisms are sometimes impractical. As most related works to this paper, a method for privacy-preserving in recursive least squares algorithm based on Secure Multi-Party Computation (SMPC) using secret sharing is proposed in [7]. Preserving the privacy of a distributed summation in a connected graph without the requirement for a trusted party is proposed in [8]. Privacy preservation in distributed optimization via dual decomposition and ADMM is another related work explored in [9], where it is desired to keep information about subsystems secret.

In this paper, we propose a distributed encrypted PID controller algorithm without a trusted party requirement. Each server holds a share of control problem which reveal nothing about private data. To this aim, distributed computing and recent result in the field of SMPC is utilized. To assess the algorithm functionality, a 2-DOF non-collocated vibrating system under earthquake excitation is considered. A schematic of the system is illustrated in Fig. 1, where dotted lines represent secure communication channels. This system can represent a 1/8 scale experimental model of a building. Sensor

is mounted on the second floor that an inertial measurement unit (IMU) is used to gather feedback signals. Let $\mathcal{F}_W = \{O; X; Y\}$ be an inertial world frame and $\mathcal{F}_B = \{o; x; y\}$ a moving frame fixed to the shake table.

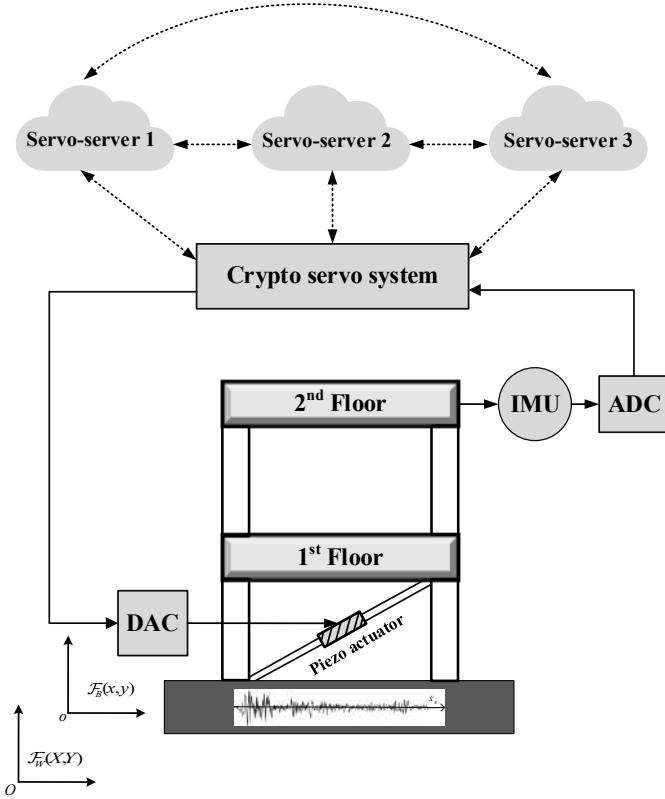


Fig. 1. The schematic of the system under control with three cloud servers and one data distributor (building).

II. PROBLEM FORMULATION

A. The Structural Modelling

We consider a 2-DOF non-collocated vibrating system under earthquake excitation as the case study illustrated in Fig. 2. After system identification using modal analysis, mathematical model of dynamic system is formulated.

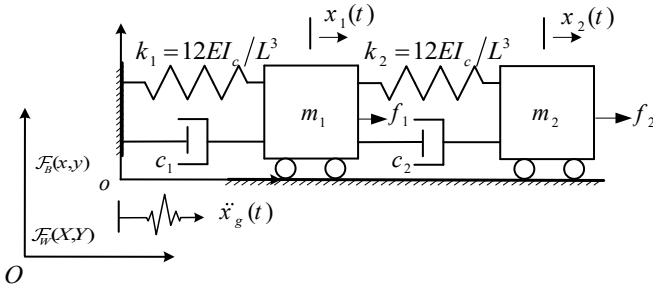


Fig. 2. Discretize model of fig.1.

This seismically excited structure can describe the dynamics of a 2-floor building shown in Fig. 3.

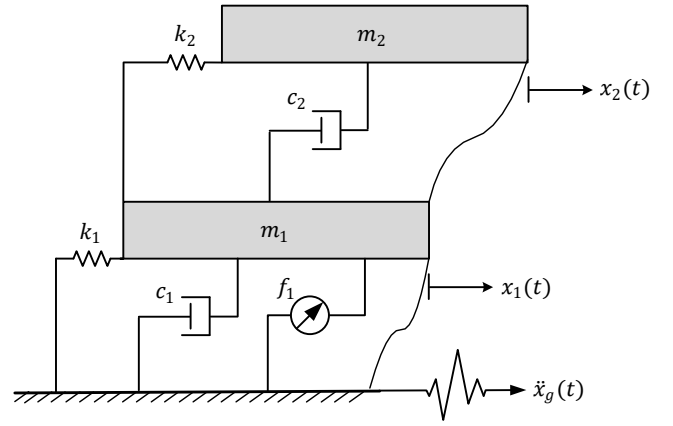


Fig. 3. Model of a 2-DOF building equipped with a control device.

However, such distributed parameter systems in civil engineering are usually complex structures, they can be simplified and idealized as discrete models as it has been depicted in fig .3. Total mass of each floor can be assumed concentrated at the top (lumped mass).

B. Analytical Formulation

The building dynamic equations of motion can be written as

$$M_s \ddot{x} + C_s \dot{x} + K_s x = f - M_s \Lambda \ddot{x}_g, \quad (1)$$

where $x = [x_1 \ x_2]^T$, $\dot{x} = [\dot{x}_1 \ \dot{x}_2]^T$, and $\ddot{x} = [\ddot{x}_1 \ \ddot{x}_2]^T$ are the vectors of displacement, velocity, and acceleration of building floors relative to the ground respectively and $f = [f_1 \ f_2]^T$ is the vector of external forces, where f_1 is the control force and $f_2 = 0$. $\Lambda = [1 \ 1]^T$ is a location vector to show the extent and distribution of excitation on each DOF [10]. M_s , C_s , and K_s are the mass, proportional damping, and stiffness matrices. For simplicity, we consider $m_{11} = 2m$, $m_{12} = m_{21} = 0$, $m_{22} = m$, $k_{11} = 2k = 24EI_c/L^3$, $k_{12} = k_{21} = -k = -12EI_c/L^3$, $k_{22} = k = 12EI/L^3$, $c_{11} = 2c$, $c_{12} = c_{21} = -c$, $c_{22} = c$. Natural frequencies and modal matrix φ of the system is obtained as follow

$$\omega_{1,2} = \sqrt{1 \pm \frac{\sqrt{2}}{2} \sqrt{\frac{k}{m}}}, \quad \varphi = \begin{bmatrix} 1 & 1 \\ \sqrt{2} & -\sqrt{2} \end{bmatrix}. \quad (2)$$

Now, from state-space representation of the system, a first-order equation is written as

$$\dot{z} = Az - B_g \ddot{x}_g + B_c f(t), \quad (3)$$

where $z = [x \ \dot{x}]^T$ is the continuous time-state variable vector of the structure [11] and

$$A = \begin{bmatrix} 0_{2 \times 2} & I_{2 \times 2} \\ -M_s^{-1}K_s & -M_s^{-1}C_s \end{bmatrix}, \quad B_g = \begin{bmatrix} 0_{2 \times 1} \\ \Lambda \end{bmatrix}, \quad B_c = \begin{bmatrix} 0_{2 \times 2} \\ -M_s^{-1} \end{bmatrix}. \quad (4)$$

Using linear transformation $x = \varphi q$, decoupled equations of motion can be formulated as

$$M \ddot{q} + C \dot{q} + K q = \bar{f}, \quad (5)$$

where $q = [q_1 \ q_2]^T$ is the vector of modal characteristics, M , C , K are modal mass, damping, and stiffness matrices, and $\bar{f}$ represents modal external forces vector.

$$M = \varphi^T M_s \varphi = \begin{bmatrix} M_1 & 0 \\ 0 & M_2 \end{bmatrix} = 4m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad (6)$$

$$C = \varphi^T C_s \varphi = \begin{bmatrix} C_1 & 0 \\ 0 & C_2 \end{bmatrix} = 8\sqrt{km} \begin{bmatrix} \xi_1 \sqrt{\frac{2-\sqrt{2}}{2}} & 0 \\ 0 & \xi_1 \sqrt{\frac{2+\sqrt{2}}{2}} \end{bmatrix}, \quad (7)$$

$$K = \varphi^T K_s \varphi = \begin{bmatrix} K_1 & 0 \\ 0 & K_2 \end{bmatrix} = k \begin{bmatrix} 4 - 2\sqrt{2} & 0 \\ 0 & 4 + 2\sqrt{2} \end{bmatrix}, \quad (8)$$

$$\ddot{f} = \varphi^T (f - M_s \Lambda \ddot{x}_g). \quad (9)$$

The modal damping arrays are calculated as $C_i = 2\xi_i \omega_i M_i$, where ξ_i is damping ratio. Now modal damping matrix is utilized to obtain system damping matrix which can be written as a linear combination of the mass and stiffness matrices

$$C_s = \theta K_s + \beta M_s. \quad (10)$$

Using (7) and (10) one obtains

$$\begin{cases} \xi_1 = \theta \sqrt{\frac{\sqrt{2}-1}{2}} \sqrt{\frac{k}{m}} + \beta \sqrt{\frac{2}{2-\sqrt{2}}} \sqrt{\frac{m}{k}} \\ \xi_2 = \theta \sqrt{\frac{\sqrt{2}+1}{2}} \sqrt{\frac{k}{m}} + \beta \sqrt{\frac{2}{2+\sqrt{2}}} \sqrt{\frac{m}{k}} \end{cases} \quad (11)$$

For $\beta = 0$ we will have a standard form of damping matrix and the best configuration is found by tuning θ . To obtain the damping matrix from the modal damping matrix, orthogonality of modal vectors with respect to the system matrices can be used

$$C_s = \varphi^{-T} C \varphi^{-1}, \quad (12)$$

where

$$\varphi^{-1} = M^{-1} \varphi^T M_s. \quad (13)$$

Now, the damping matrix can be formulated as

$$C_s = 2 \begin{bmatrix} \xi_1 \sqrt{1 - \frac{\sqrt{2}}{2}} + \xi_2 \sqrt{1 + \frac{\sqrt{2}}{2}} & \xi_1 \sqrt{1 - \frac{\sqrt{2}}{2}} - \xi_2 \sqrt{1 + \frac{\sqrt{2}}{2}} \\ \xi_1 \sqrt{1 - \frac{\sqrt{2}}{2}} - \xi_2 \sqrt{1 + \frac{\sqrt{2}}{2}} & \xi_1 \sqrt{1 - \frac{\sqrt{2}}{2}} + \xi_2 \sqrt{1 + \frac{\sqrt{2}}{2}} \end{bmatrix}. \quad (14)$$

Let $m = 50\text{kg}$, $\theta = 1.5$, $\beta = 0$, and $k = 2000\text{N/m}$. Then, the natural frequencies become $\omega_1 = 3.4228\text{ rad/s}$ and $\omega_2 = 8.2634\text{ rad/s}$, and system matrices are

$$M_s = \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix} \text{kg}, C_s = \begin{bmatrix} 252.27 & -126.14 \\ -126.14 & 126.14 \end{bmatrix} \text{Ns/m}, \\ K_s = 10^3 \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix} \text{N/m}.$$

Consequently, modal matrices are as follow,

$$M = \begin{bmatrix} 200 & 0 \\ 0 & 200 \end{bmatrix} \text{kg}, C = \begin{bmatrix} 147.8 & 0 \\ 0 & 861.3 \end{bmatrix} \text{Ns/m}, \\ K = \begin{bmatrix} 2343.2 & 0 \\ 0 & 13657 \end{bmatrix} \text{N/m}.$$

In this study, the structure model is subjected to the horizontal component of the recorded ground acceleration for the 1940 El Centro, 1978 Tabas, and 1989 Loma Prieta earthquakes depicted in Figs. 4, 5, and 6 respectively. Peak Ground Accelerations (PGA) of these earthquakes are 0.31g, 0.84g, and 0.48g respectively.

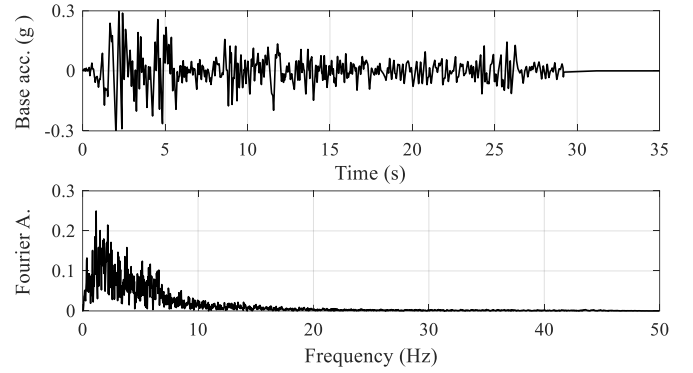


Fig. 4. Horizontal acceleration time history and DFT for El Centro earthquake (station No. 117).

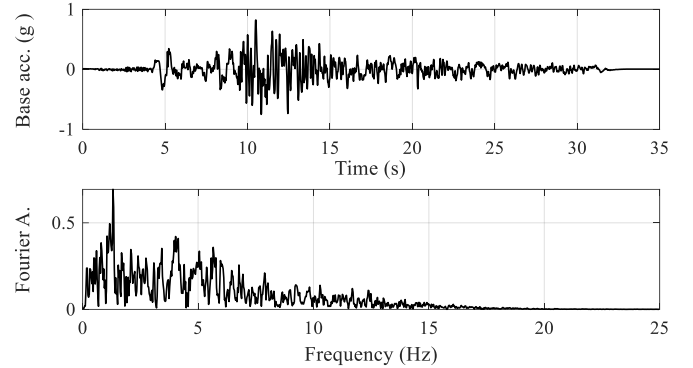


Fig. 5. Horizontal acceleration time history and DFT for Tabas earthquake (Tabas station).

III. SECURE MULTIPARTY COMPUTATION

SMPC consists of cryptographic methods and protocols which allow a set of parties (players), $\mathcal{P}_1, \dots, \mathcal{P}_n$, to hold their secret inputs $s_1, \dots, s_n$ (party $\mathcal{P}_i$ holds secret s_i). The players agree on some function g that takes n inputs. Their goal is to compute $y = g(s_1, \dots, s_n)$ while making sure that the following two conditions are satisfied; the *correctness* (the correct value of y is computed) and the *privacy* (y is the only new information that is released and each party learns nothing more than what its own input and the output of the function implied) [12]. Computing g such that correctness and privacy are achieved is referred to as computing g securely. These should be held while a specific number of parties is corrupted (a subset of parties considered to be controlled by an adversary that gathers all information that corrupted parties see during the execution of the SMPC protocol).

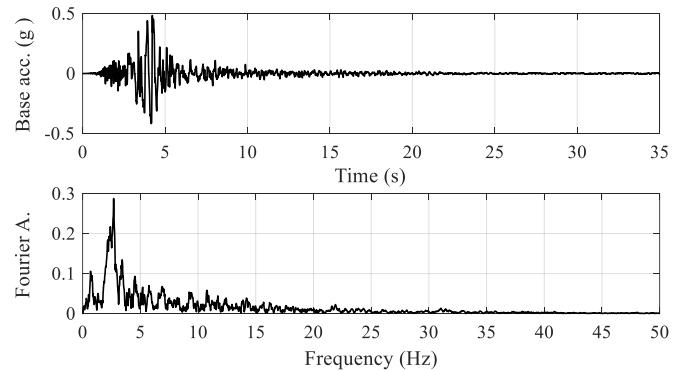


Fig. 6. Horizontal acceleration time history and DFT for Loma Prieta

One of the most fundamental results of SMPC was presented by Ben-Or and colleagues in 1988 [13]. They demonstrated that any n -party functionality can be computed with perfect security, in the private channels model. Any entity behaving maliciously is considered an *adversary*. The adversary works by taking control of a subset of the parties, which as mentioned before are referred to as corrupted parties. Adversaries can be considered *semi-honest* or *passive* (who follow the protocol specification but try to learn more than they should by inspecting the protocol transcript) and *malicious* or *active* (who may follow an arbitrary strategy attempting to manipulate the result of the computation). When the adversary is semi-honest the perfect security condition holds for less than $n/2$ corrupted parties and when the adversary is malicious this holds for less than $n/3$ corrupted parties. The complete proof and results have been studied in [14]. SMPC involves communication between parties and can include individual or hybrid approaches between techniques such as secret sharing, oblivious transfer, garbled circuits, (threshold) homomorphic encryption, etc. The SMPC protocols proposed in this paper are based on secret sharing which is a popular and frequently used encryption technique with low computational complexity.

A. Secret Sharing Scheme

Secret sharing allows a data owner to distribute its secret between a set of parties while these chunks separately reveal nothing about the secret. In fact, it takes some or all of the pieces of information to reconstruct the secret. The piece of information is referred to as a *share*. Various procedures have been suggested for secret sharing, but *Shamir's secret sharing* [15] is a well-known approach that is used in this paper. Before presenting the SMPC algorithms, it is required to introduce some notations which are used for readability of the algorithms. A τ -threshold secret sharing scheme for n parties contains two functions *Share* and *Reconstruct* where the former takes a secret s and creates n shares $s_1, \dots, s_n$ and the latter takes any set of these shares (at least $\tau + 1$ members) to produce secret s . Let $\mathcal{P}_1, \dots, \mathcal{P}_n$ be parties. Assume that there exists a secure channel between every pair of $(\mathcal{P}_i, \mathcal{P}_j)$ for $i, j = 1, \dots, n$ and $i \neq j$ and that $\{s_{i,j}\}$ for $j = 1, \dots, n$ are shares of s_i which belongs to party $\mathcal{P}_i$. When a party, $\mathcal{P}_i$, distributes the shares of s_i , it means that $\mathcal{P}_i$ securely sends $s_{i,1}$ to $\mathcal{P}_1$, $s_{i,2}$ to $\mathcal{P}_2$, and $s_{i,n}$ to party $\mathcal{P}_n$. Conversely, when a party $\mathcal{P}_i$ broadcasts a value a , it means that $\mathcal{P}_i$ makes a known to all parties. $\llbracket s \rrbracket$ represents a set of shares of s . Since, any set of τ shares or fewer cannot reconstruct the secret, the threshold can be chosen with respect to the number of expected corrupted parties. Most of the secret sharing schemes are built based on a finite field $\mathbb{F}_p = \{0, \dots, p-1\}$ where p is a large agreed prime number, and all operations are modulo p . In addition, in a secret sharing, parties are able to compute shares of the sum $\llbracket a + b \rrbracket$ and of the product $\llbracket ab \rrbracket$ modulo p , from the given shares of $\llbracket a \rrbracket$ and $\llbracket b \rrbracket$. The first property, called linearity, is achieved by most schemes without any communication requirement and is done by summing received shares of each party. The *Additive Sharing Scheme* is an example of this where the dealer draws $n - 1$ shares $s_1, \dots, s_{n-1}$ uniformly at

random over $\mathbb{F}_p$ and chooses s_n so that $s = s_1 + \dots + s_n$. $\mathcal{P}$ is a large prime number that the probability of guessing s_i by *cryptanalytic* and *brute-force* attack is negligible. Multiplication requires more communication between the parties, hence is more considerable than the linearity. This challenge can be encountered through a preprocessing phase, where a trick so-called *Beaver's circuit randomization technique* is utilized to replace distributions with broadcasting.

B. Shamir Secret Sharing Scheme (Shamir's SSS)

The underlying idea is that it takes $k + 1$ values to determine a polynomial of degree k . For instance, a linear polynomial $ax + b$ is determined by two values (a line is determined by two points) that can be done for example by Lagrange interpolation polynomials. *Shamir's SSS* uses Lagrange polynomials in order to create shares of a secret $s \in \mathbb{F}_p$. A polynomial $f(x) \in \mathbb{F}_p[s]$ of degree $\tau < n$ is chosen in such a way that $f(0) = s$ and rest of the coefficients are random. Shares of the secret s can be calculated as $s_1 = f(1), \dots, s_n = f(n)$ and distributed among n parties. Protocol 1 is proposed for creating shares of Shamir's SSS.

Protocol 1: Shares creation of Shamir's SSS	
Input:	A secret $s \in \mathbb{F}_p$ and an integer $\tau < n$, where n is number of parties
Output:	Shares $\llbracket s \rrbracket$ of s
	1. Choose coefficients $\{r_1, \dots, r_\tau\} \in \mathbb{F}_p^\tau$ uniformly at random.
	2. Define polynomial $f(x) = s + r_1x + \dots + r_\tau x^\tau$ in $\mathbb{F}_p$.
	3. Calculate shares as $s_1 = f(1), \dots, s_n = f(n)$.

After share creation, shares are distributed among parties. Suppose now that a subset of the participants tends to recover the secret. Regarding the following equation, each player owns a row of the matrix and is assigned the share corresponding to his row. This subset can reconstruct s if their rows span $< 1, 0, \dots, 0 >$, otherwise due to randomness they have no information.

$$\begin{bmatrix} 1^0 & 1^1 & 1^2 & \dots & 1^\tau \\ 2^0 & 2^1 & 2^2 & \dots & 2^\tau \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n^0 & n^1 & n^2 & \dots & n^\tau \end{bmatrix} \begin{bmatrix} s \\ r_1 \\ \vdots \\ r_\tau \end{bmatrix} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} \quad (15)$$

Protocol 2 presents output reconstruction.

Protocol 2: Output reconstruction	
Input:	Any subset $C \subseteq \{1, \dots, n\}$ of $\tau + 1$ or more shares of the $\llbracket s \rrbracket$
Output:	The reconstructed secret s
	1. Construct the recombination vector $\kappa = [\kappa_1, \dots, \kappa_n]$ via generalized Lagrange basis polynomial,
	$\kappa_i = \delta_i(0) = \prod_{j \in C, j \neq i} \frac{-j}{i-j} \quad (16)$
	2. Reconstruct the secret by polynomial,
	$f(0) = \sum_{i \in C} \kappa_i s_i \quad (17)$

s can be reconstructed by $\tau + 1$ or more of given shares via Lagrange interpolation theory. Furthermore, τ of shares reveal nothing about the secret due to the uniformly random distribution of shares and implies the privacy of *Shamir's SSS* [12]. It is obvious that the recombination vector is accessible by all parties during the execution of the MPC protocol. *Shamir's SSS* allows a function of the secret to being

computed directly on shares of secret. A general MPC protocol based on *Shamir's SSS* can be divided into three parts: *Input-sharing*, *computation*, and *output-reconstruction*. *Input-sharing* and *output-reconstruction* are similar for all protocols based on *Shamir's SSS* (Protocols 1 and 2). Hence, it is required to identify the *computation* phase for MPC protocols established in this paper. All finite functions over a finite field can be expressed as a polynomial and all needs to evaluate a polynomial are addition and multiplication. In fact, it is proved that each function $g: \mathbb{F}_p \rightarrow \mathbb{F}_p$ can be represented by a polynomial over $\mathbb{F}_p$. Protocol 3 represents an MPC for secure addition of two values a and b .

Protocol 3: Addition

Input: Parties hold shares $\llbracket a \rrbracket$ and $\llbracket b \rrbracket$

Output: $\llbracket y \rrbracket$, where $y = a + b$

1. Each party sums its received shares of a and b , i.e. $\llbracket y \rrbracket = \llbracket a \rrbracket + \llbracket b \rrbracket$.
-

This protocol can simply extend for any number of summations. Privacy of Protocol 3 is proved because only secure protocols are used. Protocol for multiplication is more complicated than for addition. Secure multiplication protocol requires $\tau < n/2$ for the *output reconstruction* to be done correctly. Protocol 4 describes the required steps.

Protocol 4: Multiplication

Input: Parties hold shares $\llbracket a \rrbracket$ and $\llbracket b \rrbracket$ where $\tau < n/2$

Output: $\llbracket y \rrbracket$, where $y = ab$

1. Each party multiplies its received shares of a and b that is, $h = \llbracket ab \rrbracket = \llbracket a \rrbracket \llbracket b \rrbracket$.
 2. Each party distributes $\llbracket h \rrbracket$.
 3. Parties compute, $\sum_{i=1}^n \kappa_i \llbracket h(i) \rrbracket = \llbracket \sum_{i=1}^n \kappa_i h(i) \rrbracket = \llbracket h_0 \rrbracket = \llbracket y \rrbracket$. (18)
-

Generally, it is very difficult to calculate the analytical solution of nonlinear differential equations and often, numerical integration of the equations is implemented [16]. Mainly, numerical integrations are divided into *explicit* and *implicit* schemes [17]. Explicit integrations are simple with conditional stability which is affected by the integration time step. In contrast, implicit integrations are unconditionally stable and can be utilized for systems with high natural frequency or multi-degree of freedom. Also, the accuracy of results can be improved by decreasing the integration time interval. The next section deals with the Newmark method as an efficient procedure with less computation than others toward capturing the dynamics of the introduced system.

IV. NEWMARK METHOD

Newmark is an implicit integration method introduced in 1959 [18]. This method is based on two assumptions

$$\dot{x}_{k+1} = \dot{x}_k + T((1 - \delta)\ddot{x}_k + \delta\ddot{x}_{k+1}), \quad (19)$$

$$x_{k+1} = x_k + T\dot{x}_k + T^2((0.5 - \alpha)\ddot{x}_k + \alpha\ddot{x}_{k+1}), \quad (20)$$

where α and δ define variations of the acceleration at a step time and specify the accuracy and stability of the method. An unconditionally stable method called constant-average-acceleration (trapezoidal rule) is obtained when $\alpha = 0.25$ and $\delta = 0.5$. In addition to (19) and (20), the following equilibrium equation is used in order to solve displacement, velocity, and acceleration.

$$M\ddot{x}_{k+1} + C\dot{x}_{k+1} + Kx_{k+1} = f_{k+1}, \quad (21)$$

where f is the vector of external forces applied to the system. Regarding the trapezoidal rule and solving for $\ddot{x}_{k+1}$, $\dot{x}_{k+1}$, and x_{k+1} , for each time step based on x_{k+1} and considering (20) and (23), one obtains

$$(4T^{-2}M + 2T^{-2}C + K)x_{k+1} = f_{k+1} + M(4T^{-2}\dot{x}_k + 2T^{-2}\ddot{x}_k + \ddot{x}_k) + C(2T^{-1}\dot{x}_k + \dot{x}_k). \quad (22)$$

Next, $\dot{x}_{k+1}$ and $\ddot{x}_{k+1}$ are obtained. The Newmark algorithm is presented in Protocol 5.

Protocol 5: Time integration Newmark scheme

Input: Parameters M , C , K , α and δ and external force f

Output: Time responses x , $\dot{x}$, and $\ddot{x}$

Initialization:

1. Form mass matrix M , damping matrix C , and stiffness matrix K .
2. Initialize $x(0)$, $\dot{x}(0)$, and $\ddot{x}(0)$.
3. Select time step T and calculate integration constants: $\delta \geq 0.5$, $\alpha \geq 0.25(0.5 + \delta)^2$, $a_0 = (\alpha T^2)^{-1}$, $a_1 = \delta(\alpha T)^{-1}$, $a_2 = (\alpha T)^{-1}$, $a_3 = 0.5\alpha^{-1} - 1$, $a_4 = \delta\alpha^{-1} - 1$, $a_5 = 0.5T(\delta\alpha^{-1} - 2)$, $a_6 = T(1 - \delta)$, $a_7 = T\delta$.
4. Calculate effective stiffness matrix, $\tilde{K}: \tilde{K} = K + a_0M + a_1C$. (23)
5. Triangularize $\tilde{K}: \tilde{K} = LDL^T$.
6. Calculate number of iterations as $N = \lfloor (T_f - T_i)/T \rfloor$, where T_i and T_f are initial and final time steps, respectively.

for $k = 1, \dots, N$ do,

1. Calculate effective load, $\hat{f}_k = f_k + M(a_0x_{k-1} + a_2x_{k-1} + a_3\ddot{x}_{k-1}) + C(a_1x_{k-1} + a_4\dot{x}_{k-1} + a_5\ddot{x}_{k-1})$. (24)

2. Solve for displacements, $LDL^T x_k = \hat{f}_k$. (25)

3. Calculate accelerations and velocities, $\ddot{x}_k(k) = a_0(x_k - x_{k-1}) - a_2\dot{x}_{k-1} - a_3\ddot{x}_{k-1}$, (26)

$$\dot{x}_k = \dot{x}_{k-1} + a_6\ddot{x}_{k-1} + a_7\ddot{x}_k. \quad (27)$$

Now the goal is to design a controller in order to reduce vibrations of the introduced system in Section II.

V. CONTROLLER DESIGN

PID controller is used broadly in the industry due to its simplicity and efficiency. The basis of this controller is a proportional term along with an integrator that leads to eliminate steady-state error and a derivative term that accelerates systems with slow time response and decreases the overshoot [19]. The output of such a controller is obtained by summing three mentioned terms

$$f(t) = K_P e(t) + K_D \dot{e}(t) + K_I \int_0^t e(\tau) d\tau, \quad (28)$$

where $f(t)$ is the control force and K_P , K_D , and K_I are the proportional, derivative, and integrator gains, respectively. Moreover, the error signal is

$$e(t) = x_{d2}(t) - x_2(t), \quad (29)$$

where $x_2(t)$ is the second floor displacement and $x_{d2}(t)$ is its desired value. The goal is to design a controller which makes the tracking error small enough. To imbed the PID controller into the Newmark one needs to discretize (28). The following relation represents the discretized PID controller at step k .

$$f_k = \left(K_P + \frac{K_D}{T}\right)e_k - \frac{K_D}{T}e_{k-1} + TK_I \sum_{i=1}^k e_i, \quad (30)$$

$$P_k = K_P e_k, I_k = I_{k-1} + TK_I e_k, D_k = K_D \frac{e_k - e_{k-1}}{T}, \quad (31)$$

where T is the time step in the Newmark method. Here, the first approximation of the backward finite difference (two-point method) is used to obtain derivative term with truncation error of $O(T)$. This can be improved by other methods such as the second approximation of the backward finite difference (three-point method) with truncation error of $O(T^2)$,

$$D_k = K_D \frac{e_{k-2} - 4e_{k-1} + 3e_k}{2T}. \quad (32)$$

Now, a closed loop system with PID controller which is integrated into the Newmark can be implemented using these formulas.

In real-world applications, system and controller are not directly connected. Indeed, output measurements of the system are gathered via sensors that are analog data. Since the digital controllers only deal with binary numbers, the analog signals have to be converted to digital signals before the control stage through *Analog to digital converters (A/D converter)*. In return, after computing the digital control signals, a *digital to analog converter (D/A converter)* is required, since the actuators work with analog signals [20].

The analog signals of the system, such as voltage, are converted to digital data in order to be readable and processed by a digital controller such as microcontrollers. ADC converts the input voltage to a binary number correspond with its size. Since the secure controller only works with integer values, the analog values in the A/D converter are quantized with a suitable scale, which will cause a quantization error. Quantization is a nonlinear effect that can result from quality constraints or internal calculations. For example, a 12-bit analog-to-digital converter converts a 10-volt range only to 4096 different values. Quantization also occurs in the multiplication of integers because multiplications are usually rounded [21]. Subtractions, to be evaluated correctly need considering negative integers in SMPC protocols. What can be done in order to consider $a < 0$ as $a \in \mathbb{F}_p$ is to examine values by the following layout in DAC unit,

$$\left\{ \underbrace{0, 1, \dots, \left\lfloor \frac{p}{2} \right\rfloor}_{\text{positive}}, \underbrace{\left\lfloor \frac{p}{2} \right\rfloor + 1, \dots, p-1}_{\text{negative}} \right\}. \quad (33)$$

As mentioned before, all calculations are performed in finite field arithmetic that has two advantages of uniform distribution of shares and preventing large unexpected values generation. Therefore, it is possible to establish a correlation between the field size and the selection of the processor power of the controller to maximize the accuracy and speed of the calculations. The method used for ADC is described in Protocol 6.

Protocol 6: A/D converter

Input: Analog signal e , desired precision σ , and finite field $\mathbb{F}_p$

Output: Shares of quantized $\llbracket e \rrbracket$

1. Truncate w with precision σ ,

$$\tilde{e} = \begin{cases} \frac{\lfloor 10^\sigma e \rfloor}{10^\sigma} & e \geq 0 \\ \frac{\lceil 10^\sigma e \rceil}{10^\sigma} & e < 0 \end{cases}. \quad (34)$$

2. Calculate and distribute scaled quantized parameter,

$$\llbracket \tilde{e} \rrbracket = \llbracket 10^\sigma \tilde{e} \rrbracket \bmod p. \quad (35)$$

The outputs of Protocol 6 are the values held by the servers. The finite field size must be chosen large enough as first the analog values are quantized in a scale and second field is halved between positive and negative integers. The used method for DAC is observed in Protocol 7.

Protocol 7: D/A converter

Input: Computed control signal $\llbracket \hat{f} \rrbracket$, desired precision σ , and finite field $\mathbb{F}_p$

Output: Analog control signal f .

1. Reconstruct $\hat{f}$, using Protocol 2.
2. For integer value $\hat{f}$

$$\begin{cases} f = \frac{\hat{f}}{10^\sigma} & \text{for } \hat{f} \leq \frac{p-1}{2} \\ f = \frac{\hat{f}-p}{10^\sigma} & \text{for } \hat{f} > \frac{p-1}{2} \end{cases} \quad (36)$$

Quantization of multiplication to be performed securely requires secure protocol of *division by public constant* to remove the additive scale of multiplication in Protocol 4 [22].

The scale of Protocol 4 inputs is halved securely by servers using Protocol 8 in order to prohibit extra scaling of multiplication. This means servers divide private analog values by $10^{0.5\sigma}$ leading to quantization error where σ is the ADC and DAC precision. Although information about the precision of the ADC and DAC units may be disclosed, this will not affect the privacy of the system.

Protocol 8: Division by public constant

Input: Parties hold $\llbracket q \rrbracket$ and public constant l

Output: $\llbracket y \rrbracket$, where $y = \lfloor ql^{-1} \rfloor$

1. Party $\mathcal{P}_1$ chooses a random value r and calculates; $r_T = \lfloor rl^{-1} \rfloor$.
 2. $\mathcal{P}_1$ distributes both r and r_T .
 3. Parties calculate $\llbracket z \rrbracket = \llbracket q \rrbracket + \llbracket r \rrbracket$.
 4. Parties open (broadcast) $\llbracket z \rrbracket$ only for party $\mathcal{P}_2$.
 5. Party $\mathcal{P}_2$ calculates $z_T = \lfloor zl^{-1} \rfloor$ and distributes $\llbracket z_T \rrbracket$.
 6. Parties calculate $\llbracket y \rrbracket = \llbracket z_T \rrbracket + \llbracket r_T \rrbracket$.
-

VI. SECURE CONTROL SOLUTION

A secure MPC protocol for distributed control can be defined using stated protocols. The block diagram of the system can be observed in Fig. 7, where double-arrows and double-framing illustrate encrypted communication and encrypted data processing, respectively.

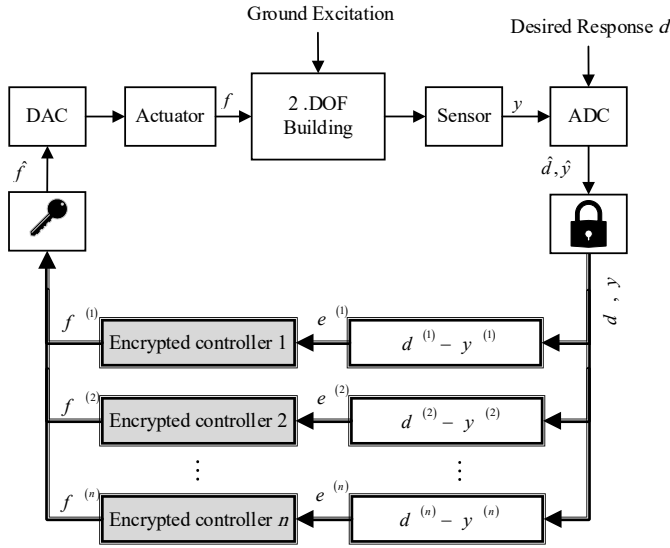


Fig. 7. Block diagram of the system with secure distributed control.

Here, n computational servers implemented in the cloud control the system securely. Also, sensor and actuator are assumed to be ideal. Given that sensor is mounted on the second mass, y_k is considered as the measured displacement of the second mass and f_k is the computed control signal at the k^{th} iteration. Regarding (30) till (32) formulas, to keep the time step of the system secure against possibly corrupted servers, dividing of shares is needed to be calculated securely although division in the finite field $\mathbb{F}_p$ is not the same as the division in $\mathbb{R}$. Despite the fact that protocols are designed to input the real numbers, an approach used to face this challenge is that replace divisions with multiplications where the inverse of the secret is distributed. The secure MPC protocol of the PID controller is stated in Protocol 9.

Protocol 9: Secure vibration control of the building

Input: Server $\mathcal{P}_i$ for $i = 1, \dots, n$ holds $\llbracket K_p \rrbracket$, $\llbracket K_D \rrbracket$, $\llbracket K_I \rrbracket$, and step time $\llbracket T \rrbracket$ and its inverse $\llbracket T^{-1} \rrbracket$, initial value $\llbracket y_0 \rrbracket$ and reference input $\llbracket d \rrbracket$

Output: Control signal $\llbracket f_k \rrbracket$ at step time k

1. Each server $\mathcal{P}_i$ for $i = 1, \dots, n$ computes $\llbracket e_k \rrbracket = \llbracket d \rrbracket - \llbracket y_k \rrbracket$.
2. Servers invoke Protocol 4 to compute $\llbracket P_k \rrbracket = \llbracket K_p \rrbracket \llbracket e_k \rrbracket$.
3. Servers invoke Protocols 3 and 4 to securely compute $\llbracket I_k \rrbracket = \llbracket T \rrbracket \llbracket K_I \rrbracket \sum_{i=1}^k \llbracket e_i \rrbracket$. (37)
4. Servers invoke Protocols 3 and 4 to securely compute $\llbracket D_k \rrbracket = 0.5 \llbracket T^{-1} \rrbracket \llbracket K_D \rrbracket (\llbracket e_{k-2} \rrbracket - 4 \llbracket e_{k-1} \rrbracket + 3 \llbracket e_k \rrbracket)$. (38)
5. Servers compute control parameter $\llbracket f_k \rrbracket$,
 $\llbracket f_k \rrbracket = \llbracket P_k \rrbracket + \llbracket I_k \rrbracket + \llbracket D_k \rrbracket$. (39)

Privacy follows from the fact that solely secure stated protocols are used. The next section represents simulations that the correctness of Protocol 9 can be observed since insecure and secure control algorithms output the same results.

VII. NUMERICAL SIMULATION

The outputs of the system under the El Centro earthquake are illustrated and corresponding results presented in the tables

is discussed. PID controller was tuned using an appropriate method for three studied earthquakes and related coefficients are given in Table I. Simulations were performed in different configurations. In the first simulation, parameters are considered as $T_f = 35$, $T = 0.008$, $n = 4$, $\tau = 1$, $d = 0$, $\sigma = 6$, and $p = 188446744073709551629$.

TABLE I. PID controller coefficients corresponding to different earthquakes.

Earthquakes	Parameters		
	K_p	K_I	K_D
El Centro	10	50	550
Tabas	100	500	700
Loma Prieta	50	450	550

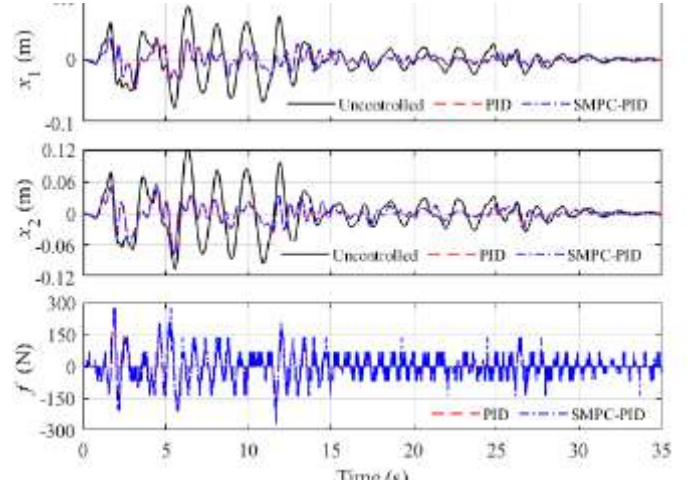


Fig. 8. Displacement of the first and second floors along with the control signal under El Centro earthquake for PID and SMPC-PID.

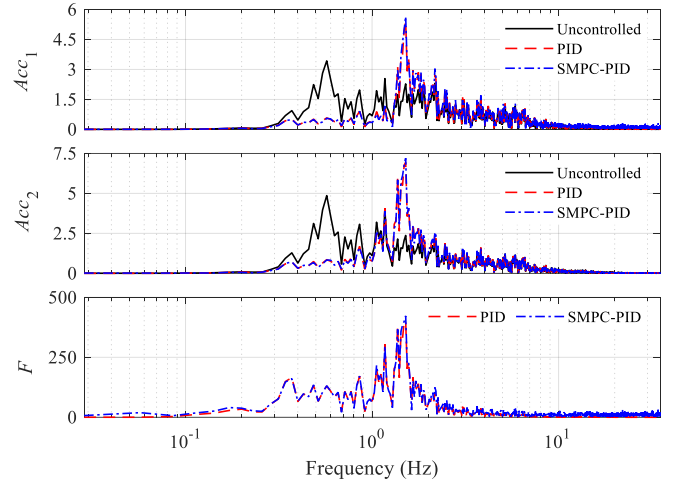


Fig. 9. DFT of acceleration of the first and second floors along with the control signal under El Centro earthquake for PID and SMPC-PID.

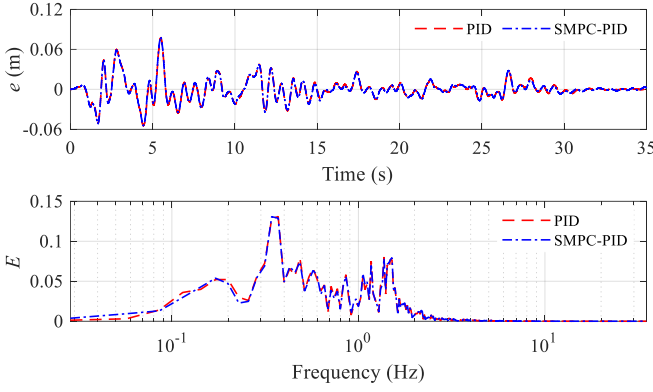


Fig. 10. Time and frequency domain error signal of system under El Centro earthquake for PID and SMPC-PID.

As it can be observed in the above figures, the results of Protocol 9 are similar to insecure PID controller responses. However, the most difference is corresponding to the control forces manifested in Fig. 8. Table II gives information about the power of the signals and also a parameter *Integral Squared Error (ISE)* is reported to compare PID and SMPC-PID algorithms.

TABLE II. Power of signals and Integral squared error of the system responses under El Centro earthquake, where $T = 0.008$, $n = 4$, $\tau = 1$, and $\sigma = 6$.

Signal	OL	CL PID	CL SMPC-PID	ISE
x_1	6.59×10^{-4}	1.142×10^{-4}	1.146×10^{-4}	1.59×10^{-6}
x_2	1.3×10^{-3}	2.522×10^{-4}	2.523×10^{-4}	2.19×10^{-6}
$\dot{x}_1$	9.02×10^{-3}	4.484×10^{-3}	4.744×10^{-3}	7.87×10^{-5}
$\dot{x}_2$	1.73×10^{-2}	9.128×10^{-3}	9.388×10^{-3}	7.57×10^{-5}
$\ddot{x}_1$	0.414	0.553	0.629	0.055
$\ddot{x}_2$	0.555	0.818	0.843	0.005
e	—	2.522×10^{-4}	2.523×10^{-4}	2.19×10^{-6}
f	—	2755	3421	521.6
$\ddot{x}_g$	0.307	—	—	—

Regarding Table II, both PID and SMPC-PID algorithms reduce displacements domain, while increasing the acceleration. Now, we solely change the A/D and D/A precision to $\sigma = 8$ and repeat the simulation which result are shown below.

TABLE III. Power of signals and ISE of the system responses under El Centro earthquake, where $\sigma = 8$.

Signal	OL	CL PID	CL SMPC-PID	ISE
x_1	6.59×10^{-4}	1.142×10^{-4}	1.142×10^{-4}	4.98×10^{-9}
x_2	1.3×10^{-3}	2.522×10^{-4}	2.421×10^{-4}	6.24×10^{-9}
$\dot{x}_1$	9.02×10^{-3}	4.484×10^{-3}	4.488×10^{-3}	2.28×10^{-7}
$\dot{x}_2$	1.73×10^{-2}	9.128×10^{-3}	9.127×10^{-3}	2×10^{-7}
$\ddot{x}_1$	0.414	0.5531	0.5538	2.15×10^{-4}
$\ddot{x}_2$	0.555	0.8177	0.8178	1.42×10^{-5}
e	—	2.522×10^{-4}	2.521×10^{-4}	6.24×10^{-9}
f	—	2755	2756	2.07
$\ddot{x}_g$	0.307	—	—	—

Obviously, by enhancing the precision parameter, SMPC-PID algorithm has much closer results to PID controller. A noticeable point is the sensitivity of control force f to the precision σ as we can observe a remarkable improvement of f in Table III that also depicted in Fig. 11.

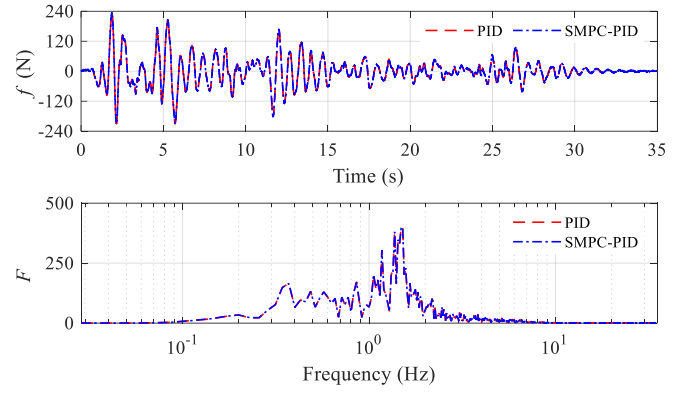


Fig. 11. Time and frequency domain control signal of system under El Centro earthquake for PID and SMPC-PID approaches, where $\sigma = 8$.

In this simulation, control action is distributed among four cloud servers with threshold 1 that each server's control signal computing performed in finite field arithmetic is shown in Fig. 12, which represents the uniform distribution of random shares.

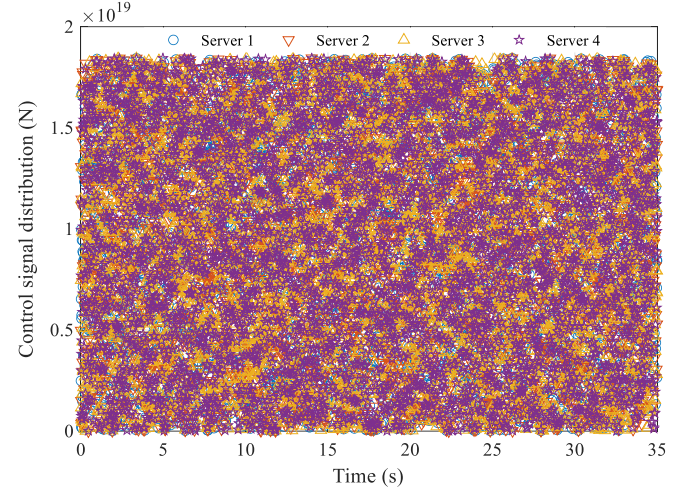


Fig. 12. Distributed control force of SMPC-PID for system under El Centro earthquake.

For Tabas and Loma Prieta earthquakes, displacements and control signal responses are depicted and entire results can be observed in tables. Desired precision of A/D and D/A is selected as $\sigma = 8$.

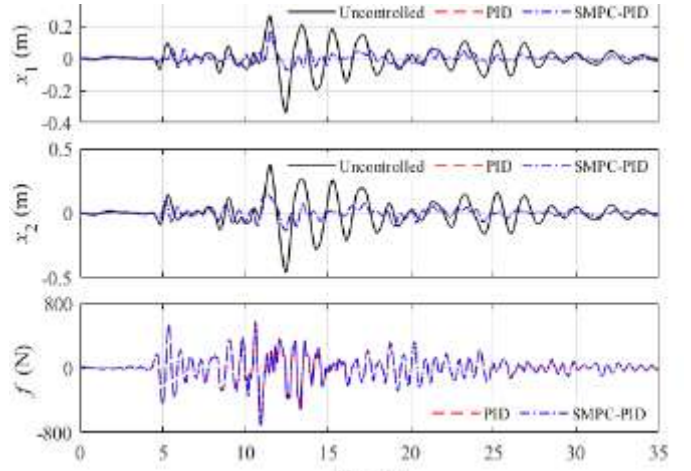


Fig. 13. Displacement of the first and second floors along with the control signal under Tabas earthquake for PID and SMPC-PID.

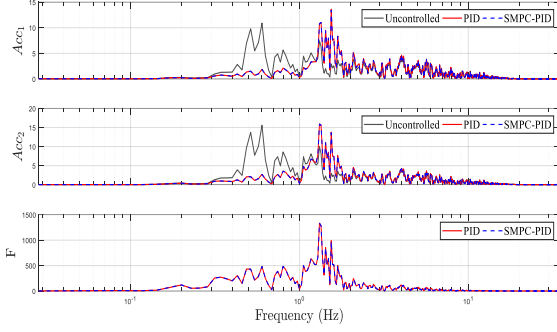


Fig. 14. DFT of acceleration of the first and second floors along with the control signal under Tabas earthquake for PID and SMPC-PID.

TABLE IV. The power of signals and integral squared error of the system responses under tabas earthquake, where $\sigma = 8$.

Signal	OL	CL PID	CL SMPC-PID	ISE
x_1	5.71×10^{-3}	7.162×10^{-4}	7.154×10^{-4}	1.2×10^{-8}
x_2	1.14×10^{-2}	1.422×10^{-3}	1.421×10^{-3}	1.3×10^{-8}
$\dot{x}_1$	7.43×10^{-2}	0.0278	0.0277	8.58×10^{-7}
$\dot{x}_2$	0.148	0.0456	0.0455	6.26×10^{-7}
$\ddot{x}_1$	2.78	3.527	3.521	3.83×10^{-4}
$\ddot{x}_2$	4.1	4.189	4.184	5.13×10^{-5}
e	—	1.422×10^{-3}	1.421×10^{-3}	1.13×10^{-8}
f	—	21480	21450	3.48
$\ddot{x}_g$	1.98	—	—	—

System responses under the Loma Prieta earthquake are depicted in the following figures and the power of signals and Integral squared error of PID and SMPC-PID approaches are stated in Table V.

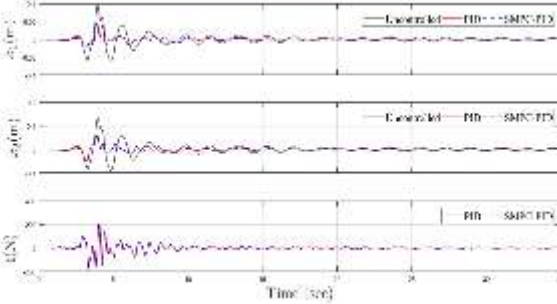


Fig. 15. Displacement of the first and second floors along with the control signal under Loma prieta earthquake for PID and SMPC-PID approaches.

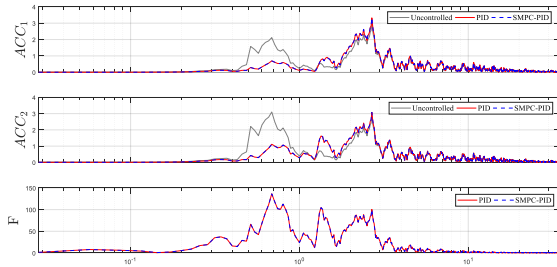


Fig. 16. DFT of acceleration of the first and second floors along with the control signal under Loma prieta earthquake for PID and SMPC-PID approaches.

TABLE V. Power of signals and Integral squared error of the system responses under Loma Prieta earthquake, where $\sigma = 8$.

Signal	OL	CL PID	CL SMPC-PID	ISE
x_1	2.18×10^{-4}	2.877×10^{-5}	2.869×10^{-5}	5.26×10^{-9}
x_2	4.44×10^{-4}	6.171×10^{-5}	6.148×10^{-5}	7.17×10^{-9}
$\dot{x}_1$	4.26×10^{-4}	1.819×10^{-3}	1.817×10^{-3}	2.68×10^{-7}
$\dot{x}_2$	7.67×10^{-3}	2.413×10^{-3}	2.409×10^{-3}	2.46×10^{-7}
$\ddot{x}_1$	0.3746	0.4141	0.4141	2.09×10^{-4}
$\ddot{x}_2$	0.4098	0.3931	0.3929	1.67×10^{-5}
e	—	6.171×10^{-5}	6.148×10^{-5}	7.17×10^{-9}
f	—	705	706.3	1.99
$\ddot{x}_g$	0.3	—	—	—

A comparison between execution time and number of servers is illustrated in Fig. 17. in spite of the fact that insecure control system has run time of 2.029 seconds. It should be pointed out that the system is under El Centro earthquake and parameters are $p = 18446744073709551629$, $T = 0.008$, $\tau = 1$, and $\sigma = 8$.

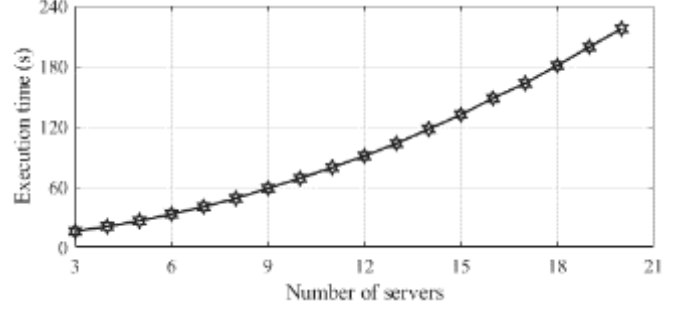


Fig. 17. Execution time versus number of servers.

VIII. CONCLUSION

The paper presents a secure multiparty protocol for a privacy preserving PID control of a 2-DOF non-collocated vibrating system under earthquake excitation. Firstly, the EOM of the system was formulated and parameters were specified using modal analysis. Next, SMPC protocols based on Shamir's secret sharing scheme were stated. The Newmark method was utilized to solve the differential EOM of the introduced system. Digital control strategy integrated into the Newmark method was stated. ADC and DAC units were introduced to communicate the physical layer and distributed controllers. After that, the proposed secure control algorithm was introduced and privacy was discussed. finally, simulations depicted similar results with the insecure algorithm. Despite the fact that it was observed SMPC-PID control signal has more sensitivity to ADC precision, it doesn't affect system responses considerably.

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